

DATA/CS 172: Python for Data Analysis

For Loops

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Announcements

- Lab 3 due Wednesday

Agenda

- For Loops
 - For working with files
 - Fixed number of repetitions

Right now...

Two kinds of scripts:
Run top to bottom



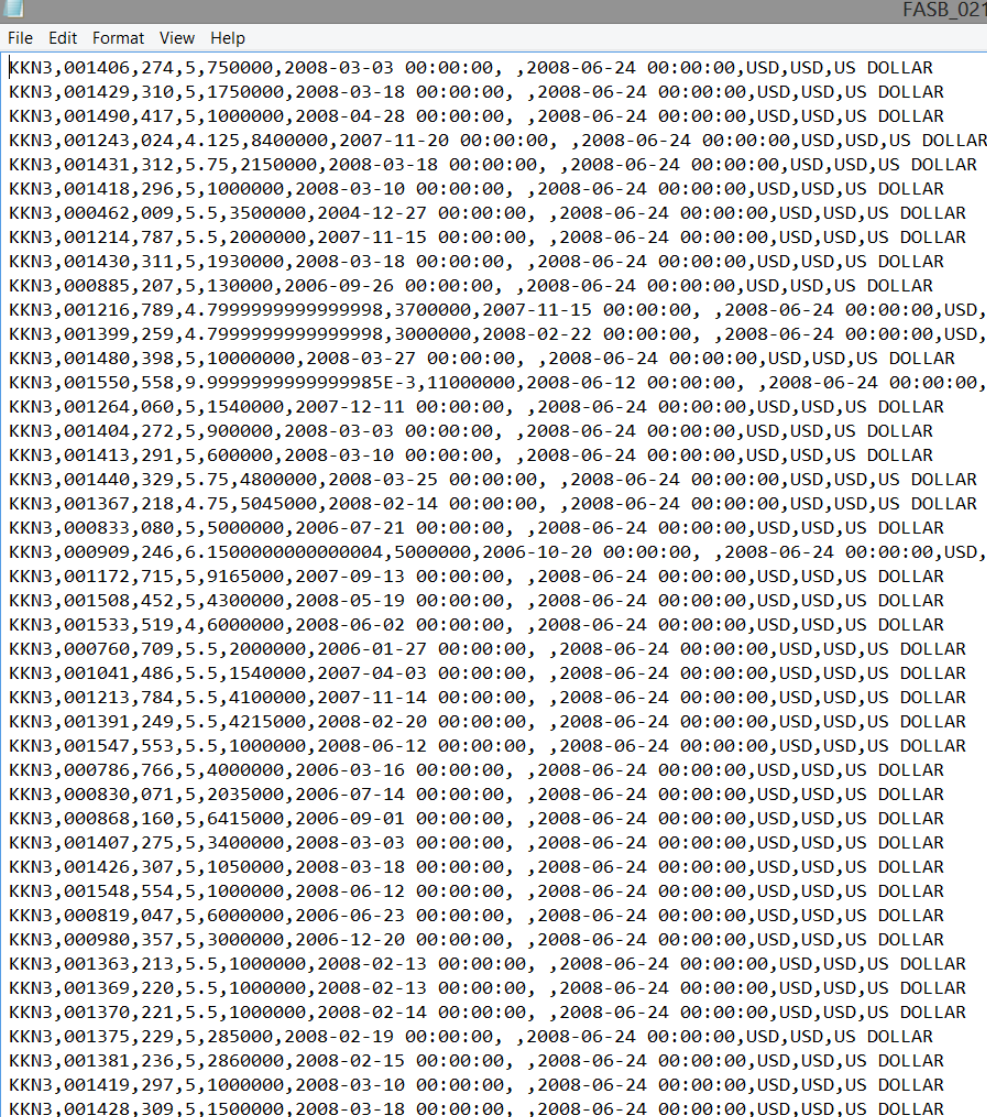
Branch with if statements



Can do some cool stuff, but something feels missing...

In data science...

- Usually you have a file with a lot of datapoints in it
- Why have all our files so far just had **one thing**????
- It's because we don't have any way to **repeat** code!
 - So if I had a file with 100 datapoints I would have to copy & paste my code 100 times!
 - And good luck if I later decide to change something!

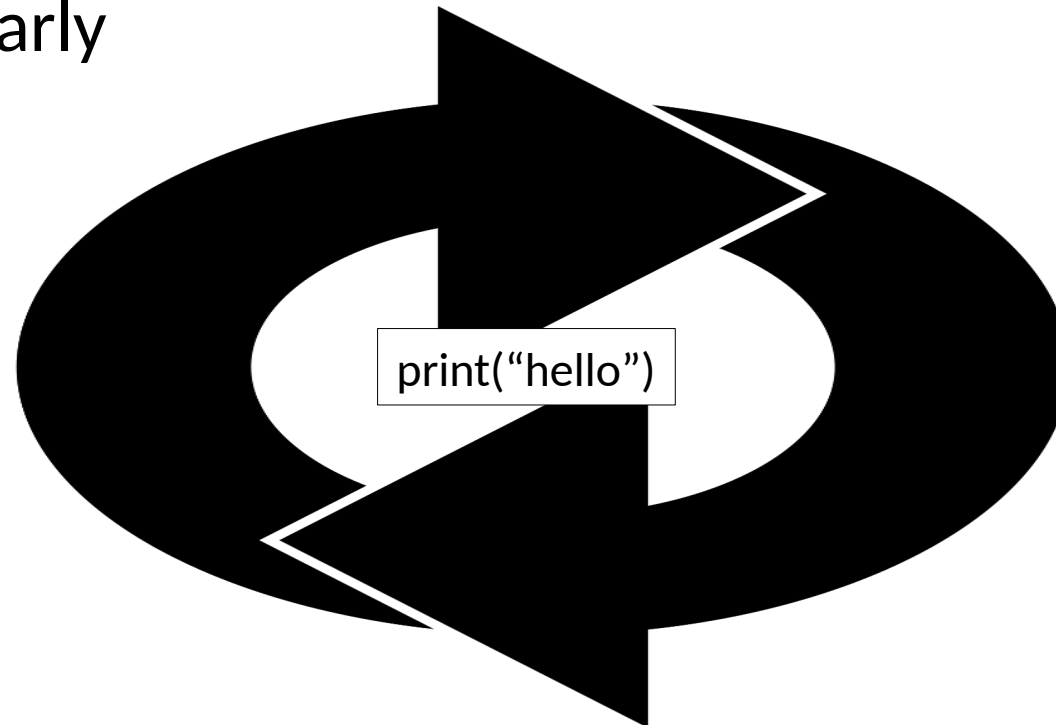


The screenshot shows a text editor window with a menu bar (File, Edit, Format, View, Help) and a title bar (FASB_021). The text area contains a long list of repetitive data points, each line starting with 'KKN3' followed by various numerical values and dates, all ending with ',USD,USD,US DOLLAR'. The text is highlighted in blue, indicating it is selected.

```
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KKN3,001214,787,5.5,2000000,2007-11-15 00:00:00, ,2008-06-24 00:00:00,USD,USD,US DOLLAR
KKN3,001430,311,5,1930000,2008-03-18 00:00:00, ,2008-06-24 00:00:00,USD,USD,US DOLLAR
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KKN3,001428,309,5,1500000,2008-03-18 00:00:00, ,2008-06-24 00:00:00,USD,USD,US DOLLAR
```

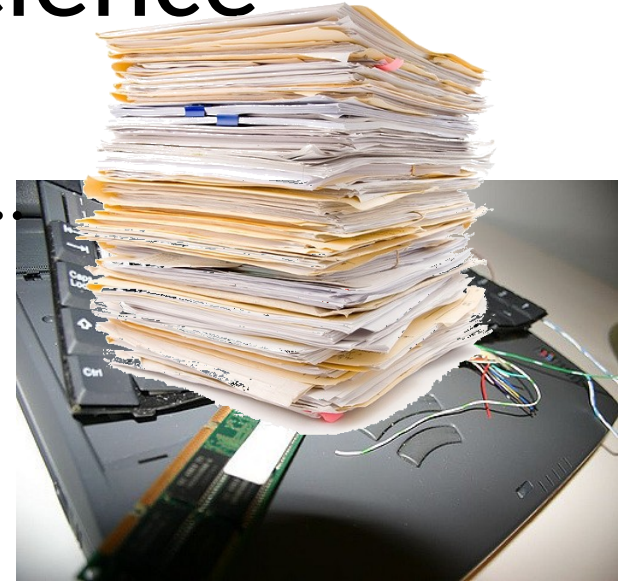
Loops

- A **loop** let us repeats any chunk of code over and over and over.... ← **important term!**
 - Single line, or multiple lines
- Way better than retyping
- Excellent for data science when there is whole bunch of data we want to process similarly



The Most Important Loop in Data Science

- As a data scientist, you will often be working with files...
- So far we've just been using `read()`
 - Which reads the entire file into a **string**!
 - What if the file has a lot of info?
 - It's going to be slow/a memory hog
 - And a headache to get all of that info out of our giant string!
- So in data science, the **#1** most important loop is the loop that goes through all the lines in a file!
 - At least for text files... we will see similar things with other types of files later though



Looping through lines in Python

`f = open("filename.txt", "r")` ← Hopefully this is familiar!

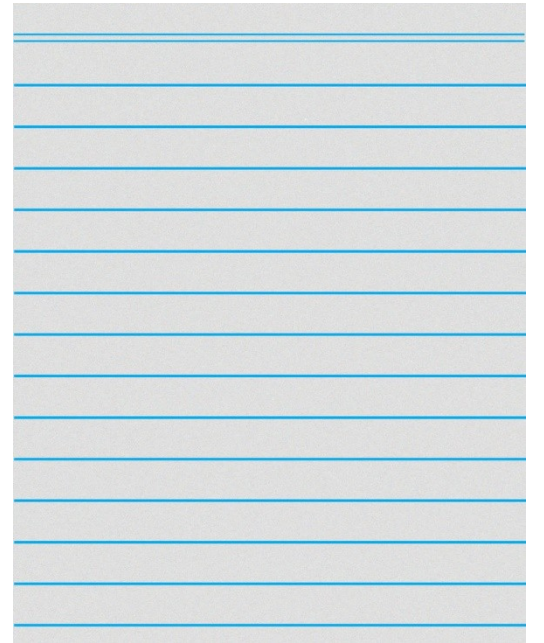
for line in f:

`print("Line:", line)`

`#Still inside for loop`

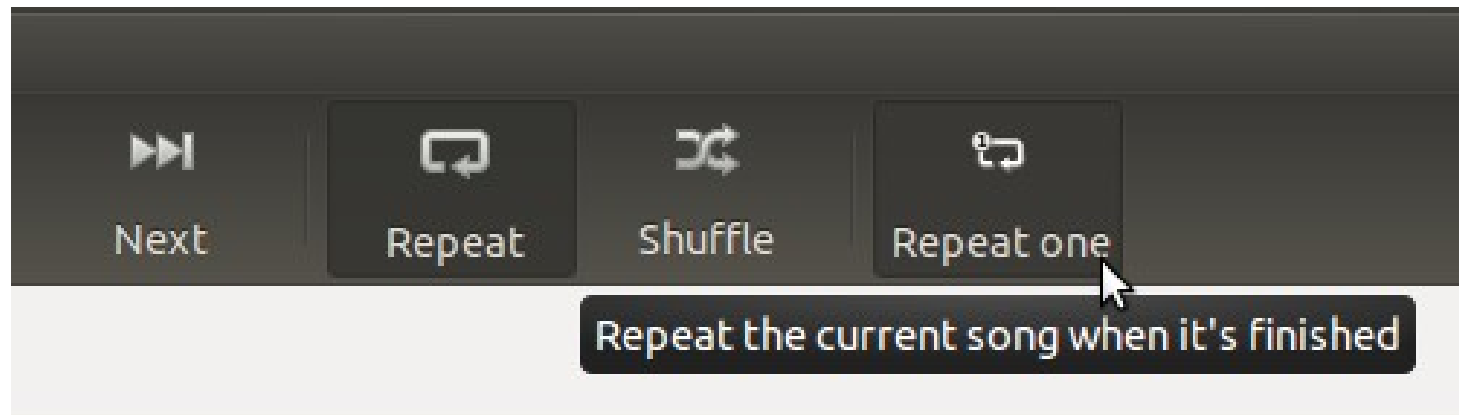
`#outside for loop`

- Special loop in Python
 - Doesn't read whole file into to memory, just reads a line at a time
- 99% of the time this is how people read text files



Is this the only kind of loop we need?

- No!
- Sometimes you want to do something twice, or three times, or four times...
 - For example, go through 3 files, 4 files, etc.
 - Being able to repeat code comes in really handy!



Fixed iteration for loop

```
for i in range(5):  
    print(i)
```

prints

0

1

2

3

4

Range with a start

```
for i in range(7,10):
```

```
    print(i)
```

7

8

9

- Comes in handy when you want to skip some initial stuff
- **Remember:** Inclusive of first endpoint, exclusive of last!

Lab out

- Getting started with loops